AGE 310: Instrumentation and Measurements

Dr. Kadeghe Fue, Department of Agricultural Engineering, SUA

Morogoro, Tanzania

Abstract

Lecture 11 introduces the fundamental concepts of feedback control. We establish the anatomy of control loops, contrasting the simplicity of Open Loop control with the accuracy and complexity of Closed Loop control. The lecture details common control strategies, including On-Off control (with hysteresis) and the industrial standard: Proportional-Integral-Derivative (PID) control, which is essential for stabilizing processes like tractor guidance and chemical rate application.

Lecture 11: Introduction to Control Systems

11.1 Fundamentals of Control

11.1.1 Control System Definition

A **Control System** is a collection of components that manages, directs, or regulates the behavior of other devices or systems to achieve a desired output. In agricultural engineering, control systems manage variables like temperature, soil moisture, engine speed, or fertilizer flow rate.

- **Setpoint (SP):** The desired value (e.g., $28°$C in a greenhouse).
- **Process Variable (PV):** The actual measured value from the sensor (e.g., $26.5°$C).
- **Error (e):** The difference between the Setpoint and the Process Variable.

$$e = \text{SP} - \text{PV}$$

  The controller's job is to drive the error to zero.

11.1.2 Open Loop Control

The control action is **independent** of the output. The system applies a predefined action for a set duration, assuming the desired result will be achieved.

- **Structure:** Input $\rightarrow$ Controller $\rightarrow$ Actuator $\rightarrow$ Process $\rightarrow$ Output. (No feedback path).
- **Example:** A timer-based irrigation system that waters for 15 minutes every morning, regardless of recent rainfall or current soil moisture.
- **Pros:** Simple, cheap, inherently stable.
- **Cons:** Inaccurate, unable to compensate for disturbances (e.g., strong wind cooling the greenhouse).

11.2 Closed Loop Control (Feedback)

11.2.1 Closed Loop Principle

The control action depends directly on the output, which is measured and fed back to the controller. This allows the system to self-correct.

- **Structure:** Input $\rightarrow$ Comparator (Error Summing) $\rightarrow$ Controller $\rightarrow$ Actuator $\rightarrow$ Process $\rightarrow$ Sensor $\rightarrow$ Comparator.
- **Example:** A moisture-sensor-based irrigation system. The pump runs until the soil moisture sensor detects the SP has been reached.
- **Pros:** High accuracy, automatically compensates for external disturbances, highly adaptable.
- **Cons:** More complex, requires a sensor, and can be unstable (oscillate) if not properly tuned.

11.2.2 Components of the Feedback Loop

1. **Sensor (Measurement):** Converts PV (e.g., temperature) into a measurable electrical signal (PV_m).
2. **Comparator (Σ):** Calculates the Error ($e = SP - PV_m$).
3. **Controller (Brain):** Calculates the corrective action based on the error.
4. **Actuator (Muscle):** Executes the command (e.g., turns on the pump).
5. **Plant/Process:** The physical system being controlled.

11.3 Basic Control Modes

11.3.1 On-Off Control (Two-Position)

This is the simplest feedback mechanism. The actuator has only two states (ON or OFF).

- **Logic:** If PV $<$ SP, actuator is ON (100% power). If PV $>$ SP, actuator is OFF (0% power).
- **Limitation:** Leads to constant switching or *chatter* near the setpoint, severely damaging relays and motors.

11.3.2 On-Off with Hysteresis

To prevent chatter, a deadband (Hysteresis) is introduced between the ON and OFF points.

- **Example (SP $= 25^\circ$C):** Heater turns ON at 23°C and turns OFF at 27°C.
- **Effect:** Reduces actuator cycling but causes the PV to continuously oscillate (ripple) around the SP. The magnitude of oscillation depends on the system's inertia (thermal mass).

Calculation Example 1: Hysteresis Band

A controller manages the fertilizer mixing process, maintaining the pH setpoint at 6.5. To prevent chatter, a 0.4 unit Hysteresis is applied.

1. Calculate the ON threshold (acid pump activation).
2. Calculate the OFF threshold (acid pump deactivation).

Solution:

- Hysteresis Band: 0.4 pH units (meaning ± 0.2 pH from SP).
- ON Threshold (PV too high, actuator reduces pH): SP $+ 0.2 = 6.5 + 0.2 = 6.7$ pH (This assumes the actuator adds base/alkaline, or the control switches logic, but for simple hysteresis based on SP).
- Standard interpretation: Actuator runs until SP $+ 0.2$ and stops until SP $- 0.2$. Let's assume the actuator adds base when pH drops too low.
- Actuator ON (if pH falls): $6.5 - 0.2 = 6.3$ pH
- Actuator OFF (when pH is restored): $6.5 + 0.2 = 6.7$ pH

The system cycles between 6.3 pH and 6.7 pH.

11.4 PID Control (The Industrial Standard)

The Proportional-Integral-Derivative (PID) controller is the most commonly used algorithm in industrial control, allowing precise, stable, and fast regulation. The final control output $u(t)$ (Actuator power) is the sum of three terms.

11.4.1 Proportional Term (P)

The output is proportional to the current error (e).

$$u_{\text{p}}(t) = K_p \cdot e(t)$$

- **Function:** Provides immediate reaction. The bigger the error, the bigger the corrective push.
- **Limitation:** Leads to a permanent **Steady-State Error** (Offset). A small error is never completely eliminated because the output goes to zero as the error approaches zero.

11.4.2 Integral Term (I)

The output is proportional to the accumulation (sum) of past errors.

$$u_{\text{i}}(t) = K_i \int e(t)dt$$

- **Function:** Eliminates the steady-state error (offset). If a small error persists, the integral term slowly builds up until it forces the PV exactly to the SP.
- **Limitation:** Slows down the response and can cause *Integral Wind-up* (overshoot) if the system is saturated (max power).

11.4.3 Derivative Term (D)

The output is proportional to the rate of change (slope) of the error.

$$u_{\text{d}}(t) = K_d \frac{de(t)}{dt}$$

- **Function:** Acts as a brake. It anticipates future error by looking at the current trend. Used to suppress overshoot and dampen oscillations.
- **Limitation:** Amplifies high-frequency noise from the sensor, potentially causing chatter in the actuator. Hence, filtering is crucial (Lecture 6).

11.4.4 The Full PID Equation

The total output to the actuator:

$$u(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt}$$

Tuning: The process of selecting the correct values for K_p, K_i, and K_d to achieve the desired dynamic response (e.g., critically damped). Methods include Ziegler-Nichols (based on ultimate gain/period) or iterative manual tuning.

11.5 Agricultural Control Applications

11.5.1 Rate Control for Sprayers

Goal: Maintain chemical application rate (e.g., 100 L/ha) constant, regardless of tractor speed (V).

- **Inputs:** Speed Sensor (GPS or Radar), Flow Meter (PV).
- **Actuator:** Proportional Control Valve.
- **Control:** PID calculates the required valve opening based on the error between actual flow and target flow (which depends on V). If V increases, the valve opens to increase flow, ensuring the L/ha ratio remains constant.

11.5.2 Tractor Hitch Draft Control

- **Objective:** Maintain consistent tillage depth by regulating the hydraulic hitch height.
- **Sensor:** Load cell (Strain Gauge) measures the pulling force (*draft*) on the implement.
- **Control:** If draft increases (e.g., hits hard ground), the PID controller uses the hydraulic cylinder (Actuator) to lift the implement slightly, reducing the draft and stabilizing the load on the engine.

Calculation Example 2: P-Control Offset

A Proportional-only controller ($K_p = 2$) controls a heater. The system requires 40% power output to maintain the setpoint temperature (SP) of 30°C due to heat loss. When the temperature stabilizes, the measured PV is 28°C. Calculate the *Steady State Error* (Offset).

Solution:

- The final controller output u must be 40% or 0.4.
- The final error e is required to generate this output using the P-term:

$$u = K_p \cdot e \implies 0.4 = 2 \cdot e$$

$$e = 0.4/2 = 0.2^\circ\text{C}$$

- The controller can only generate 40% power when there is a 0.2°C error.
- PV $=$ SP $- e = 30^\circ\text{C} - 0.2^\circ\text{C} = 29.8^\circ\text{C}$
- The Steady State Error is 0.2°C. This constant offset is the defining limitation of P-only control.

11.6 Summary of Key Terms (Swahili/Context)

- **Control System:** Mfumo wa Udhibiti (System that regulates output).
- **Setpoint (SP):** Thamani Inayohitajika (Desired target value).
- **Process Variable (PV):** Thamani Halisi Inayopimwa (Actual measured value).
- **Closed Loop:** Mzunguko wa Mrejesho (Feedback control).
- **Hysteresis:** Upinzani wa Mabadiliko (Deadband to prevent chatter).
- **PID:** Udhibiti wa PID (Proportional, Integral, Derivative control).
- **Integral Wind-up:** Mzigo wa Integral (Overshoot caused by accumulated error).
- **Steady-State Error:** Hitilafu ya Kudumu (Permanent offset in P-control).

Lecture 11: Review Questions

SECTION A: Basic Concepts (5 Questions)

1. Define *Setpoint* (SP) and *Process Variable* (PV). How is the *Error* calculated in a feedback control system?
2. Differentiate between *Open Loop* control and *Closed Loop* control. Give one advantage and one disadvantage of the Open Loop system.
3. Explain the problem of *chatter* in simple On-Off control. How is the principle of *Hysteresis* used to solve this problem?
4. State the primary function of the Proportional (P) term in a PID controller. What is the main drawback of using P-only control?
5. Describe the function of the Integral (I) term. How does it resolve the limitation of the P-only control mode?

SECTION B: Advanced Analysis and Calculations (10 Questions)

6. **Control System Stability (Dynamic Link):** A heating system (First Order, $\tau = 15$ s) is controlled by an On-Off controller with 4°C Hysteresis band. If the setpoint is 30°C, sketch the temperature response over time. Explain how increasing the system's thermal mass (e.g., adding more water to the heating volume) affects the oscillation frequency and amplitude.
7. **PID Tuning Philosophy:** Explain the distinct roles of the K_p, K_i, and K_d gains in shaping the *Step Response* of a system (Speed, Offset, and Overshoot).
8. **Integral Wind-up Mitigation:** Describe the phenomenon of *Integral Wind-up*. Give one common method used in professional PID control software (PLCs) to prevent this when the actuator is saturated (running at 100% power).
9. **Hydraulic Draft Control Logic:** In the context of tractor draft control using hydraulic cylinders, if the sensor (Load Cell) reports a sudden *decrease* in draft force, what instantaneous action should the PID controller command the hydraulic valve to do, and why?
10. **Rate Control Calculation:** A chemical rate controller needs to spray at 150 L/ha. The tractor is currently traveling at 2 m/s over a 10 m boom width. Calculate the required instantaneous flow rate (Q) in L/min. (ha $= 10,000$ m^2).
11. **Derivative Term Limitation:** Explain why the Derivative (D) term is typically calculated based on the *rate of change of the PV* rather than the *rate of change of the Error* when the setpoint is subject to sudden changes (Setpoint Jump).
12. **Control Loop Identification:** Identify the components (Sensor, Controller, Actuator, Plant) in the following system: A geothermal greenhouse uses a PT100 to measure soil temperature, a Raspberry Pi to execute a Python script, and a pump to circulate warm water through underground pipes.

13. **P-Control Design:** A proportional controller controls the speed of a grain conveyor motor ($0 - 100\%$). The desired speed is 50 RPM (SP). At $K_p = 0.5$, when the speed stabilizes at 48 RPM, what output power percentage is the controller supplying to the motor?

14. **Pneumatic Actuator Precision:** A simple pneumatic cylinder is controlled by an On-Off valve. Explain why this system is unsuitable for high-precision, continuous motion control, relating your answer to the physical property of the fluid (air).

15. **Control System Selection:** Recommend the most suitable control mode (On-Off vs. PID) for the following applications and justify your choice:

    - Controlling a simple water pump based on a float switch in a storage tank.
    - Stabilizing the altitude of an agricultural spraying drone (UAV).